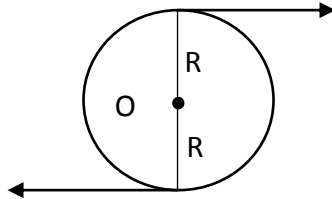


➤ **Torque or twisting moment (T)**

If an element with its length along x axis goes moment along M_y or M_z , it bends in XZ & XY planes. However, when moment along x-axis is applied, it results in twisting of section. Such a moment is called torque.

In other words, the moment of couple acting on the shaft is called torque.



Torque = twisting force x diameter of shaft

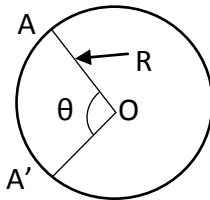
$$T = F \times 2R$$

T = Torque (N.mm)

R = radius of shaft

F = turning force

➤ **Angle of twist (θ): -**



When a shaft is subjected to torque, point A on the surface of the shaft comes to A' position. The angle AOA' at center of the shaft is called the angle of twist.

Angle AOA' = θ = angle of twist (radians)

➤ **Strength of shaft: -**

Maximum torque or power the shaft can transmit from one pulley or other, is called strength of shaft.

(a) **Solid Circular shaft**

$$T = \frac{\pi}{16} \tau D^3$$

D = diameter of the shaft

τ = shear stress in shaft

(b) **Hollow circular shaft**

$$T = \frac{\pi}{16} \tau \left(\frac{D^4 - d^4}{D} \right)$$

D = outer diameter of the shaft

d = inner diameter of the shaft

➤ **Shear stress in shaft (τ)**

When a shaft is subjected to equal & opposite end couples, whose axis coincides with the axis of the shaft, the shaft is to be in pure torsion and at any point in the section of the shaft stress will be induced that stress is called shear stress in shaft.

➤ **Polar moment of inertial (J)**

The moment of inertial of a plane area with respect to an axis perpendicular to the plane of the figure is called polar moment of inertia.

➤ **Polar section modulus (Z_p)**

It is a ratio of polar moment of inertia to distance of extreme to centroid

$$Z_p = \frac{J}{Y} = \frac{J}{R}$$

(a) **Solid circular shaft**

$$Z_p = \frac{\pi}{16} D^3$$

(b) **Hollow circular shaft**

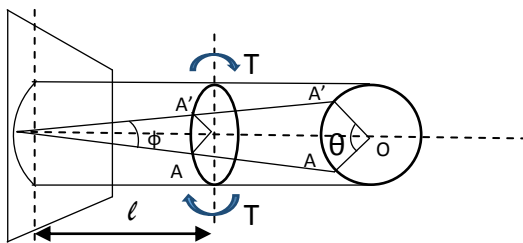
$$Z_p = \frac{\pi}{16D} (D^4 - d^4)$$

➤ **Assumptions in the theory of torsion**

The following assumptions are made while finding out shear stress in a circular shaft subjected to torsion.

- 1) The material of shaft is uniform through-out the length.
- 2) The twisting along the shaft is uniform
- 3) The shaft is of uniform circular section the through-out the length
- 4) Cross-sections of the shaft, which are plane before twist remains plane after twist.
- 5) All radius which are straight before twist remains straight after twist.

➤ **Theory of torsion & Torsion equation.**



- Consider a shaft fixed at one end & subjected to torque at the other end.

Let,

T = torque

l = length of shaft

R = radius of the shaft

- As a result of torque every cross-section of the shaft will be subjected to shear stresses.
- Line CA on the surface of the shaft will be deformed to CA' & OA to OA' as shown in figure.

Let,

Angle $ACA' = \phi$ in degree = shear strain

Angle $AOA' = \theta$ in radian = angle of twist

τ = shear stress induced at the outermost surface of the shaft.

C = modulus of rigidity.

Now, shear strain = $\frac{AA'}{l} = \tan \phi = \phi$ (it is very small)

Length of arc $AA' = R.\theta$

$$\phi = \frac{AA'}{l} = \frac{R.\theta}{l} \dots \dots \dots (1)$$

Modulus of rigidity (c) = $\frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\tau}{\phi}$

$$\phi = \frac{\tau}{C} \dots \dots \dots (2)$$

Now, equating 1 & 2,

$$\frac{\tau}{C} = \frac{R.\theta}{l}$$

$$\frac{\tau}{R} = \frac{C.\theta}{l} \dots \dots \dots (3)$$

We know that for solid circular shaft,

$$T = \frac{\pi}{16} \tau D^3$$

$$\tau = \frac{16 \cdot T}{\pi \cdot D^3}$$

Substitute the value of τ in equation 3.

$$\frac{\frac{16T}{\pi \cdot D^3}}{R} = \frac{C \cdot \theta}{l} \dots \dots \dots (R = \frac{D}{2})$$

$$\frac{T}{\frac{\pi}{32 \cdot D^4}} = \frac{C \cdot \theta}{l}$$

$$\frac{T}{J} = \frac{C \cdot \theta}{l} \dots \dots \dots$$

4

From equation 3 & 4

$$\frac{T}{J} = \frac{\tau}{R} = \frac{C \cdot \theta}{l} \dots \dots \dots \text{Equation of Torsion.}$$

➤ **Power transmitted by a shaft: -**

(a) Power in horse power (h.p)

$$P = \frac{2\pi NT}{4500} \text{ h.p}$$

Where, N = round per minute (R.P.M)

T = Torque in Kg.m

(b) Power in watts

$$P = \frac{2\pi NT}{60} \text{ watt}$$

1 Kw = 1000 watt

1 h.p = 746 watt

1 h.p = 0.746 Kw.

➤ **Torsional Rigidity**

Let twisting moment T, produce a twist θ radian in a length l.

As we know that $\frac{T}{J} = \frac{C \cdot \theta}{l}$ so that $\theta = \frac{T \cdot l}{C \cdot J}$

- For a given shaft the twist is therefore proportional to the twisting moment T.
- In a beam the bending moment produce deflection, in the same manner a torque produces a twist in a shaft.
- The quantity CJ in the above equation known as a torsional rigidity. This quantity stands for the torque required to produce a twist of 1 radian per unit length of the shaft. It is also corresponding to similar quantity EI, in expression for deflection of beams. EI is known as flexural rigidity.
- The quantity CJ/l is known as torsional Stiffness
- The Quantity l/CJ is known as torsional Flexibility.

Example: 1 A 40 mm diameter steel shaft of length 1200mm is used to transmit 50 kw between a motor and a pump. Determine the lowest speed of rotation at which shear stress does not exceed 60 MPa and angle of twist does not exceed 2°

$$D = 40 \text{ mm}$$

$$L = 1200 \text{ mm}$$

$$P = 50 \text{ Kw}$$

$$\tau = 60 \text{ MPa}$$

$$\theta = 2^\circ = (2 \times 3.14)/180 = 0.0349 \text{ radians}$$

For Stiffness

$$C = 80$$

$$\tau = 60 \text{ MPa}$$

$$\theta = 0.0349 \text{ radians}$$

$$\frac{T}{J} = \frac{C \cdot \theta}{L}$$

$$\frac{T}{251327.41} = \frac{80 \times 1000 \times 0.0349}{1200}$$

$$T = 584755.11 \text{ N.mm}$$

For Strength

$$T = \frac{\pi}{16} \tau D^3$$

$$= (3.14/16) \times 60 \times 40^3$$

$$= 753982.23 \text{ N.mm} = 753.98 \text{ N.m}$$

$$P = \frac{2\pi NT}{60}$$

$$N = (50 \times 10^3 \times 60 \times 2 \times 3.14 \times 753.98)$$

$$N = 633.26 \text{ rpm}$$

$$J = \frac{\pi}{32} D^4 = 251327.41 \text{ mm}^4$$

Power:

$$P = \frac{2\pi NT}{60}$$

$$50 \times 10^3 = \frac{2\pi \times N \times 584.76}{60}$$

$$N = 816.51 \text{ rpm}$$

The lowest speed N = 633.26 rpm

Example: 2 Calculate the diameter of the shaft required to transmit 45 kW at 120 rpm. The maximum torque is likely to exceed the mean by 30 %, for a maximum permissible shear stress of 55 N/ mm². Calculate also the angle of twist for a length of 2m. G = 80 x10³ N / mm²

$$P = 45 \text{ Kw} = 45 \times 1000 \text{ watt}$$

$$\tau = 55 \text{ N/mm}^2$$

$$N = 120 \text{ rpm}$$

$$l = 2000 \text{ mm}$$

Maximum Torque

$$T_{\max} = 1.30 \times T_{\text{mean}}$$

$$= 1.30 \times 3580.98$$

$$= 4655.28 \text{ N.m}$$

$$T_{\max} = 4655.28 \times 10^3$$

Angle of Twist

$$\frac{\tau}{R} = \frac{G \cdot \theta}{l}$$

$$\theta = 0.0364 \text{ radians}$$

Mean torque

$$P = \frac{2\pi NT}{60}$$

$$45 \times 10^3 = \frac{2\pi \times 120 \times T}{60}$$

$$T = 3580.98 \text{ N.m}$$

Diameter of Shaft

$$T = \frac{\pi}{16} \tau D^3$$

$$4655.28 \times 10^3 = (3.14/16) \times 55 \times D^3$$

$$D = 75.54 \text{ mm}$$

Example: 3 Calculate diameter of shaft to transmit 10 Kw at a speed of 15 Hz. The maximum shear should not exceed 60 MPa.

$$P = 10 \text{ Kw} = 10 \times 1000 \text{ watt}$$

$$\tau = 60 \text{ N/mm}^2 (\text{MPa})$$

$$N = 15 \text{ Hz} = 15 \text{ Cycle/second} = 15 \times 60 \text{ rpm} \\ = 900 \text{ rpm}$$

Torque:

$$P = \frac{2\pi NT}{60}$$

$$10 \times 10^3 = \frac{2\pi \times 900 \times T}{60}$$

$$T = 106.10 \text{ N.m}$$

$$T = 106.10 \times 10^3 \text{ N.mm}$$

Diameter of Shaft

$$T = \frac{\pi}{16} \tau D^3$$

$$106.10 \times 10^3 = (3.14/16) \times 60 \times D^3$$

$$D = 20.80 \text{ mm}$$

Example: 4 A shaft has to transmit 105 Kw power at 160 rpm. If the shear stress is not to exceed 65 N/mm² & the twist in a length of 3.5 m must not to exceed 1°. Find suitable diameter. Take $G = 8 \times 10^4 \text{ N/mm}^2$

$$P = 105 \text{ Kw} = 105 \times 1000 \text{ watt}$$

$$\tau = 65 \text{ N/mm}^2 (\text{MPa})$$

$$N = 160 \text{ rpm}$$

$$\theta = 1^\circ = (1 \times 3.14)/180 = 0.0174 \text{ radians}$$

Torque in Shaft

$$P = \frac{2\pi NT}{60}$$

$$105 \times 10^3 = \frac{2\pi \times 160 \times T}{60}$$

$$T = 6266.72 \text{ N.m}$$

Now let's find out the diameter of the shaft based on its strength & stiffness

For Strength:

$$T = \frac{\pi}{16} \tau D^3$$

$$6266.72 \times 10^3 = (3.14/16) \times 65 \times D^3$$

$$D = 78.89 \text{ mm}$$

Provide greater of two values of D

$$D = 112.55 \text{ mm}$$

For Stiffness

$$J = \frac{\pi}{32} D^4$$

$$15.756 \times 10^6 = (3.14/32) \times D^4$$

$$D = 112.55 \text{ mm}$$

Exmple: 5 A solid steel shaft has to transmit 120 Kw at 600 rpm. Find the diameter of the shaft if the shear stress is to be limited to 100 N/ mm². Estimate the possible % saving in the material of the shaft if hollow shaft of internal diameter equals 0.75 times external diameter is replaced against solid shaft.

$$P = 120 \text{ Kw} = 120 \times 1000 \text{ watt}$$

$$\tau = 100 \text{ N/mm}^2 (\text{MPa})$$

$$N = 600 \text{ rpm}$$

Diameter of Solid Shaft

$$P = \frac{2\pi NT}{60}$$

$$120 \times 10^3 = \frac{2\pi \times 600 \times T}{60}$$

$$T = 1909.85 \times 10^3 \text{ N.mm}$$

$$T = \frac{\pi}{16} \tau D^3$$

$$1909.85 \times 10^3 = (3.14/16) \times 100 \times D^3$$

$$D = 45.98 \text{ mm}$$

Replacing Shaft

$$T = \frac{\pi}{16} \tau \left(\frac{D^4 - d^4}{D} \right)$$

$$1909.85 \times 10^3 = \frac{\pi}{16} \times 100 \times \left(\frac{D^4 - (0.75)^4 D^4}{D} \right)$$

$$= \frac{\pi}{16} \times 100 \times (0.6836 D^3)$$

$$D = 52.20 \text{ mm}$$

$$d = 0.75 D$$

$$= 39.15 \text{ mm}$$

% Saving in material

$$A_{\text{solid}} = \frac{\pi}{4} \times 45.98^2 = 1660.45 \text{ mm}^2$$

$$A_{\text{hollow}} = \frac{\pi}{4} \times (52.20^2 - 39.15^2) \\ = 936.28 \text{ mm}^2$$

% Saving in material,

$$= \frac{A_1 - A_2}{A_1} \times 100$$

$$= \frac{(1660.45 - 936.28) \times 100}{1660.45}$$

$$= 43.61 \%$$

Hence, % saving in material is 43.61 %.